

Library Inventory Management System

Name: Mehek Pramod Sawant

Roll no: 34

Class/Batch: SYIT- 02

Project Title :Library Inventory Management System

1. Project Description:

The **Library Inventory Management System** is a Python-based desktop application developed using Tkinter to manage library book records. Its purpose is to make it easier to add, search, and delete book information while keeping the records organized. Users can enter the book title, ISBN/Book ID, genre, and availability status through a simple graphical interface. The records are displayed in a table and saved in a CSV file, with different colors used to identify available and borrowed books.

2. Code Snippet:

```
def add_book():
    data = (
        len(book_list) + 1,
        title_entry.get().strip(),
        isbn_entry.get().strip(),
        genre_combo.get(),
        status_combo.get()
    )
    if not all(data[1:]):
        return messagebox.showwarning("Warning", "Please complete all fields!")

    book_list.append(data)
    insert_row(data)

    with open("library_inventory.csv", "a", newline="", encoding="utf-8") as f:
        csv.writer(f).writerow(data)

    messagebox.showinfo("Success", "Record successfully added!")
    clear_fields()

def insert_row(book):
    tag = "available" if book[4] == "Available" else "borrowed"
    table.insert("", "end", values=book, tags=(tag,))

def search_book():
    query = isbn_entry.get().strip().lower()
    table.delete(*table.get_children())

    if query in ("", "all"):
        matches = book_list
    else:
        matches = [b for b in book_list if str(b[0]) == query or query in b[1].lower() or query in b[2].lower()]

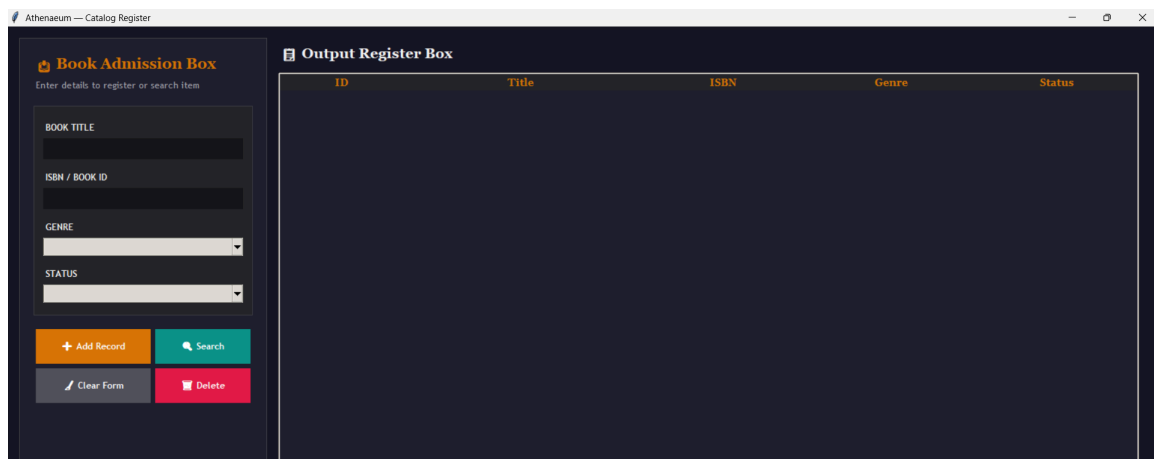
    if not matches:
        return messagebox.showerror("Not Found", "No matching book found!")

    for b in matches:
        insert_row(b)

def delete_book():
    selected = table.selection()
    if not selected:
        return messagebox.showwarning("Warning", "Please select a record to delete!")

    for item in selected:
        table.delete(item)
    messagebox.showinfo("Success", "Selected record deleted!")
```

3. Output Screenshot:



4. Video Demonstration:

https://drive.google.com/file/d/1k1Ku7b9cYiQLO-d_i7yRuHQlBxjg7yCj/view?usp=drive_link

 Screen Recording 2026-09-29 110015.mp4

5. GitHub Repository:

<https://github.com/mehekps216-debug/Python-Mini-Project-library-/blob/a6b40fa212024b348e20f7b5ab06e9328c9b8f0e/mini%20project.py>

6. Additional Notes:

During the development of the project, challenges included managing the Tkinter GUI components, handling user input, and storing records correctly in a CSV file. The project helped in understanding **Python GUI development, functions, lists, file handling, CSV operations, and event-driven programming**. In the future, the system can be improved by adding features such as **book issue/return management, database integration, login authentication, and automatic due-date tracking**.